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Evaluating Innovative Financing Mechanisms for the California High-Speed Rail Project

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Abstract:

Millions of dollars are involved in high-speed rail (HSR) infrastructure construction and maintenance. Large-scale projects like HSR require funding from a variety of avenues beyond those available through public monies. Although HSR serves the general public’s mobility needs, any funds (whether State or Federal) flowing from the public exchequer usually undergo strict review and scrutiny. Funds from public agencies are always limited, making such traditional financing mechanisms unsustainable for fulfilling HSR’s long-term operational and maintenance cost needs—on top of initial costs

involved in construction. Therefore, any sustainable means of financing HSR projects would always be welcome. This research presents an alternate revenue generation mechanism that could be sustainable for financing HSR’s construction, operation, and maintenance. The methodology involves determining key HSR stations, which, after development and improvement, could significantly add value to businesses and real estate growth. Any form of real estate taxes levied on properties surrounding such stations could substantially support the HSR project’s funding needs. In this research, a bi-objective optimization problem is posed in conjunction with a Pareto-optimal front framework to identify those key stations. With 28 California HSR stations used as an example, it was observed that the four proposed HSR stations in Fullerton, Millbrae-SFO, San Francisco Transbay Terminal, and San Diego would be excellent candidates for development. Their development could increase the economic vitality of surrounding businesses. The findings could serve as valuable information for California HSR authorities to focus on developing key stations that would generate an alternate funding source for an HSR project facing funding challenges.

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